Some thoughts about presentations (drawn mostly from *The Chicago Guide to Communicating Science,* by Scott L. Montgomery)

The spoken word is the music of knowledge: momentary, personal, expressive.

Scientific journals began in the 17th century as collections of talks that had been given in Paris or London and revised after criticism and discussion. "Writing and speaking were inseparable, like color and taste in an excellent wine."

But listening and reading are different. Remember: "Speaking is dynamic, transient, and reliant on human interaction." Dull and monotonic recitations are an insult to an audience.

Montgomery's/Reid's basic advice on preparation:

1. *Begin preparing your talk well in advance.* Outline the structure, consider alternatives, think about visuals.
2. *Design and write out your talk* in a manner you feel comfortable with. Using presentation software is common, but be thoughtful about how you construct your slides. You are telling a story, not simply delivering bullet points and figures. Be prepared to rearrange your material—and when you do, be careful to make sure that the slides form a coherent narrative.
3. *Practice your talk* when it is ready. Montgomery recommends doing so at least 10 times. (Who has that kind of time?) Do this by yourself and in front of a test audience if you have one. If you're going to give a "big talk," find an occasion to give a version of the talk to a smaller audience, and insist on candid feedback about content, delivery, everything.
4. When you practice, *pace yourself.* A good rate of speed is about 90–120 words per minute. This means about 2,000 words for a 20-minute talk. These rates are significantly slower than conversational speech. Measured speech allows the audience to keep up with you, and it gives you the appearance of control and fluidity.
5. *Stage fright?* Consider it an asset. Under control, adrenaline and nervousness provide energy that can be channeled into the talk. Breathe slowly; visualize yourself doing well. Generally, each talk is easier than the last!
6. *Plan to interact and adapt.* Unless the talk is formal, with no questions allowed, plan for moments of informal interaction. Plan the talk flexibly enough that it can be adjusted as you judge the audience's level of interest and comfort. Questions may indicate a need for additional explanation. And you will have to deal with the clock. Remember that you don't need to fill up all the time, and your allotted time may change.

Structure of a professional talk:

1. Why you did this work

2. How you did it
3. What your findings were
4. What it means

Fine structure:

Introduction: In a professional talk, you often begin by acknowledging colleagues and funders, thanking organizers, other courtesies. Humor is always good. In a public talk, it's great to be able to start on a humorous note. Metaphors and metaphorical images are good for connecting the audience to the topics you will speak about.

Remember: People always need some time to settle into their seats. A good speaker can use the opening to get use to an audience and establish a relationship.

Outline: It is good to give a sense of what you're going to cover, but this need not be extensive. Ideally the same terms used in the outline should be repeated as each topic arrives later.

Aristotelian formula: Have a beginning, a middle, and an end! The middle is the core of a talk. Often talks have several "middles"—a great deal of detail packed in—so that they never find their end. The details need to be trimmed out of the middle so that you can make a logical progression to the end.

Some standard problems in presentations today:

1. *Too many slides*.
2. *Breezing by key ideas.* On a chalkboard, what is written persists, helping the listener connect ideas. Slide presentations can put extra demands on the listener because fundamental ideas, equations etc. must be held in the mind without visual support. Assertive listeners often demand that the speaker repeat previous slides. Be thoughtful about reinforcing and connecting concepts.
3. *Information clutter.* It is very easy to put far too much on slides, making it hard for listeners to follow the simple development of an idea.
4. *Visual clutter.* Graphics exist, and therefore are overused. The signal-to-noise ratio can be poor in presentations that use an excess of animated effects, backgrounds, junky graphics and a rainbow palette. Less is more!
5. *Word clutter.* Again, less is more. Even when a speaker is ready to move on, the audience can remain busy trying to read wordy slides. A talk is not a paper! (Are the slides to be posted online? Perhaps extra words, pictures and explanatory slides can be put in the online version and left out of the actual presentation.)
6. *The next-slide problem.* Computerized slide talks often have moments where the presenter clicks on the next slide, then stops to inspect the slide

before deciding where the talk is going next. A skillful presenter does this discreetly and without breaking the flow of the talk. A poor presenter tends to forget where he/she was going next, so that the talk seems continually to wander. Other presenters bring inappropriately large slide sets, so that they must make multiple leaps, skipping slides, to get to the end of the talk. These problems are moderated by practice and preparation and by having podium notes or a printed slide set or outline.

7. *The word "so."* This verbal crutch is a word that actually has a meaning, roughly "therefore." When used sparingly, it can create a sense of one thing following another. I maintain that a speaker who restrains the overuse of "so" shows a higher level of confidence in what is being said. "So" is such a common filler word that *not* using it can have quite an effect.
8. *Giving a talk to and for oneself.* Some talks seem created with the speaker as an audience, rather than being written with the listener's viewpoint and interests in mind. A talk is given for an audience; its principal objective is to engage and satisfy that audience.
9. *Movies.* Occasionally someone launches a movie without having explained what will be seen. If possible, a movie should be preceded by a static graphic or a couple of snapshots explaining what will be shown, what to look for and important parameters such as color scheme and temporal and spatial scales.

Considering aesthetics and information in creating transparencies and slides:

Slide layout principles:

1. English is read from left to right. Most people will understand a graphic better if explanatory text appears on the left and the graphic on the right.
2. Label things clearly! Often graphics are inserted into a slide from a file created by a data-analysis package. Axis labels might be missing, or might be expressed in symbols or terms not understandable to the audience. They might be written sideways on a vertical axis, so that the audience will have to tilt their heads to read them. When possible, create simple but clear labels for data in your slides. These will stand in for the captions used in a publication. This step is worth whatever time it takes. It will keep you from having to stop to explain the axes or to back up to answer questions about your graphs.
3. Choose a slide background based on whether your presentation will be graphics-heavy or text-heavy; also, if possible, consider the light environment in which you will speak. If there is a lot of text, it is easiest on the eyes to use dark lettering on a white or light-colored background. Graphics such as animations are more easily seen against a dark-colored background. Choose one background, since it is hard on the eyes to switch back and forth. A talk given in a completely dark room often works better if the background is dark.

4. Information content: Assume that a slide will be viewed for one minute or less. Look at each slide and see whether you can get oriented to its content in 20 seconds or so and can understand it fully in another 30–40 seconds. If it requires more attention, consider moving some content to another slide.

Preparing talks for new audiences:

Cardinal Rule: Look at your slides and expect to revise them every time. It is almost always obvious to an audience when a talk has been prepared for a different audience. Check the fine points: Are my axes labeled with symbols that will be confusing to this audience?

McCormick/Reid tips:

1. For all but the most specialized audiences, prepare didactic slides that can be inserted at the beginning and at the introduction of new concepts. Make sure to use these in any interdisciplinary or interspecialty setting.
2. For every step moving from specialist toward lay audiences, strip out equations and carefully examine data and terminology to make sure there are no terms or measurements that will not be clear to the new audience.
3. Again, at each step, slow down the pace at which new ideas are introduced. Allow time for comprehension. Pause. Especially when things get technically challenging, plan bits of humor or fun that will help maintain the audience's comfort level and provide time for things to sink in.
4. Enthusiasm is also more important at each step away from the specialized environment of your own subdiscipline. Enthusiasm will convince the skeptical that your work has importance and will convey the excitement of science to unfamiliar audiences.
5. A talk for an academic audience broader than your discipline is much like a talk to undergraduates. Beware the urge to impress your lab colleagues; ditch the jargon. Be sure to practice this talk on someone outside your department.
6. Solicit honest feedback every time you give a talk to a new kind of audience. Use this feedback in refining what you do the next time. You will get better and better, but only by listening to your audiences.
7. Stunts are more and more effective the broader the audience. Especially if they're humorous, or done in a self-effacing manner, they will relax the audience and build a connection between speaker and audience, and they will hold interest. Audience-participation stunts appeal to younger groups.
8. Use the feedback from undergraduate teaching to sort out what works in public talks. What makes their eyes glaze over? What makes them smile and sit up?

A few personal observations:

Every speaker has a different approach. Find your own. I find that I must work through a talk by writing out all I would like to say, then selecting the major points, then putting together my slides—and that all this preparation allows me to ad lib the actual talk. But this is not what I recommend to others; others will find that very different approaches work better for them

Identify your weaknesses. Mine are nervousness and a tendency to talk far too fast. I found this out through video- and audiotaping. Whenever you give a talk that is recorded, watch carefully and ask for feedback from others.

Interact with meeting organizers as much as possible. You need to find out about your audience, about the purpose of the meeting or occasion and your role in it. Otherwise you might give a beautiful speech that is utterly wrong for the occasion and audience.

Herbert Hirsch's words of wisdom:

The keys to successful presentation are *poise, confidence* and *dignity.*

Poise: Speak well and clearly at a controlled pace. Gesture appropriately and interact comfortably with whatever audience you're handed.

Confidence: Confidence breeds poise. Planning and preparation breed confidence.

Dignity: The unforeseen will happen. Audiences will be hostile, computers will malfunction. Dignity will carry you through.

References:

Montgomery, Scott L., *The Chicago Guide to Communicating Science.* University of Chicago Press, 2003.

Hirsch, Herbert L., Essential Communication Strategies for Scientists, Engineers, and Technology Professionals, Second Edition. IEEE Press, 2003.